NIST-Post Quantum Cryptography Algorithm Optimization and Scarcity

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***Abstract*—In 2022-2027 NIST science and technology organization of quantum cryptography that estimates the challenges in computing and optimizing the data filtration in the form qubit scalability. This has been the challenging in the hardware and loading the classical data in the form of qubits to make the platform flexible and high computation rates in executing or loading the code. In the scenarios of the quantum cryptographic algorithm and the data scarcity in computing and load the data. The main issue in implementing the data is in the data synchronization and the complexity of computation. In this technique to enhance the flexibility and the scalability in computing the classical data by using the machine learning algorithm i.e. QNN (quantum neural network) and Shor's & Grover's algorithm etc. this process can lead you to be the data integration and the data categorization in the form of qubits. This paper computes the data with the help of quantum algorithm and make the strategical analysis in the making the architectural framework. This paper shows the quantum algorithmic architectural framework analysis approach to finds the data scarcity and the optimization of the data computing in data-set. The aim is to find the classical survey analysis on quantum cryptography issues estimates to find the flexibility and quantum analysis of qubits.**

Keywords—NIST cryptography, Security Analysis, classical data, quantum cryptography, machine learning, qubits, QNN, quantum learning

I. Introduction

The NIST Quantum cryptography is the phenomenon approach by which it estimates the nature of the quantum analysis in the era of structure analysis and the techniques of the en-cryption and the de-cryption of the computing analysis in classical Qubit form[1]. This type of estimation can be taken in term oof the computing strategies that can be used in formulating and making the outcome procedure in terms of the malware analysis in the computing and also makes the data optimization free from the attack[2,3]. This also recognizing the strategies of the Quantum approach in finding the big algorithmic approach in quantum cryptography[4]. It also plays an important role in Quantum Cryptography area by making and authenticating each and every parameter of the algorithmic approach in the authorized and unauthorized analysis of the data in terms of the Qubit. The Qubits can be taken in the form of the digit 0 and 1 [5,6]. The NIST is the organization which can used to make the fully optimization and data scarcity approach to finalized the algorithm based on the requirement analysis. It act as a barriers of the security act happen in computing the data in the data-set to provide the flexibility and security in computing[7]. There are few algorithm approach in Quantum approach are[7]:

1. CRYSTAL-Kyber (it is the mechanism that can secure the data in the form of Qubit and the Quantum analysis)
2. CRYSTAL-Dilithium (it is DSA scheme that can prevent the attack in the data set)
3. Sphincs+ (based on their hash algorithm that can be used to prevent it from malware attack)
4. Falcon (it can be used as a DSA algorithm to fight against from the security attack)

The techniques that can be taken based on their four Cryptographic approach algorithm that can be taken in terms based on the 2022-23 [8]. The upcoming updated Falcon algorithm and it can be coming in around 2024-25 and with new updating and enhanced the security[9]. All these algorithm based on to secure the data from the malware attack in the quantum crypto analysis in the classical data-set [10]. The digital signature algorithm is implemented in terms of the encryption and decryption techniques by using the authentication and authorization of the key to access and make the classical analysis in the data analysis [11]. The post quantum security analysis attack in the form of the KEM (key encryption mechanism) that can be used in the lattice-based analysis for the mathematical analysis in the form of Qubit forms. All these algorithm can be used for the cryptographic analysis and it can be prevented from the security attack like-CRYSTAL-Kyber, CRYSTAL-Dilithium, Sphincs+, Falcon algorithm. Accordingly, QNN learning approach to enhance its data optimization and data scarcity in term of the classical data Qubit form 0 and 1 [12].

Shors Cryptographic Algorithm[12]: It is the Quantum Algorithm that can be implemented in the terms of the integer form that can be implanted in the terms of the Qubits form and finds the prime factors of every analysis of the Quantum approach. It is mainly used to boost the flexibility and the scalability by using shor;'s algorithm. The time and space complexity is o(logn) and can used as a classical analysis of each parameters.

B^{r} mod N =1 (It can be uses to factorized each and every parameter)

Grover's Algorithm [13] : It is the phenomenon approach that can be implemented in the form of the boosting and speeding up the sorted and unsorted of the classical dataset Qubits algorithm that can enhances the computing strategies and make the measurement of the super-position of each parameter. This can be used in the form of the cryptographic analysis, quantum data optimization & the data scarcity in the requirement terms of the multiple forms to execute the data analysis.

$A^{n} = 2|0^{n}> <0^{n}| - K_{n}$.............Grover's operator o(√n) times.

QNN (Quantum Network) [14] : Quantum Neural network is the phenomenon algorithm by which it can take every parameter as an input detail of the data to make the hidden analysis and the corrections off the data parameter in the form the each and every Qubit form to make the formulations and the calculation of the each parameter and the correction of the error in hidden layer approach. The process of authentication and the make the analysis of the Qubit formulation and also take the analysis goes to meet the each and every parameter. The data can be taken as an input layer and goes in the hidden layer approach to detect the error and corrected at the layer wise approach and give you mathematical analysis parameter in the data-set. This algorithm concept follows the neural network approach to identify each parameter equation 1,2,3,4,5 .[14,15].

{A} = cos B + I sin C = e^{a}...(1)

G = ∑A (B1).X1- A(€)= ∑A (B1).A(C1)-A(€)……..(2)

C= 3.14 h (μ)-argument(G)…………………………(3)

Hidden layer (J)=A(B)………………………………(4)

W &w1 , A,X= quantum phase…………………………(5)

A= function and h= activation function

h (μ)= 1/1+e μ for every input parameter network

This algorithmic approach is the high functionality statement that can be implemented in the terms of the high computation and the high hidden layer approach in the dataset [16,20]. The basic approach is to make estimation of the hidden layer approach to make the estimation of the simulation technique to all hidden layer approach..

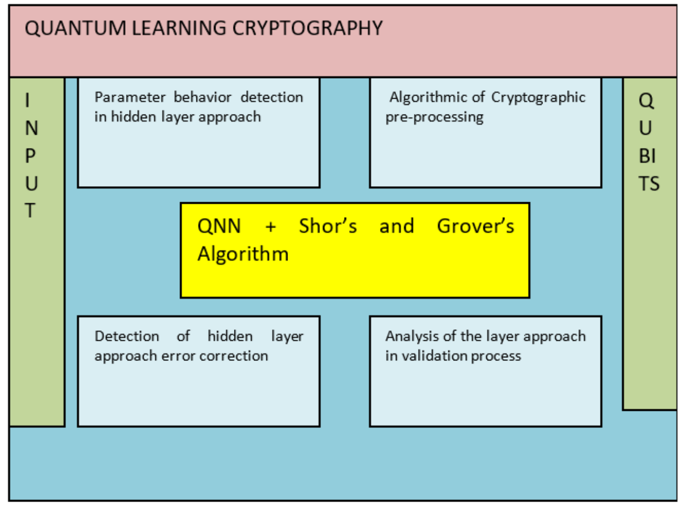

Fig. 1. Quantum cryptographic analysis

This paper estimates the analysis of the cryptographic analysis on the various dataset by implementing the data in the form of Qubits and the analysis of the processing techniques that can estimates analysis of Qubits of Quantum cryptographic analysis . This algorithmic analysis can be taken to enhances the flexibility and scalability and the optimization while computing the data in the data-set and the data scarcity.

II. Literature Survey

[1,2,3] The NIST post-Quantum approach is the systematic approach by which it can be translated structured terms of analysis of computing the data in the DSA cryptographic algorithm and the analysis of the transitional approach of the NIST analysis. In truncation process of transitional approach by which it can be communicated based on their cryptographic algorithmic approach in the scenario of algorithmic analysis. The vast form of the strategies analysis in computing and making the structural algorithmic analysis and free from the computing strategies.

[4,5,6] the Quantum Navigation techniques is the strategies by which it can be communicated in terms of the structural way of organizing and making the Algorithmic approach in the DSA Algorithm and the various environment related to the cryptographic algorithm in terms of the Qubits forma and the making the Strategies analysis of the structural form of the digits in the form of 0 and 1. The efficient analysis of the NIST algorithmic approach is can estimated the way of the functional approach of the simulation of the techniques in terms of the accuracy and free from the security attack while computing the data in terms of the data-set. [7,8,9] this paper estimates the whole computing cryptographic algorithmic approach into one approach to make t the strategies analysis in the terms of the Qubits form and make the process analysis based on the environment analysis of the structural analysis. This can lead to the challenges of the structural analysis. The empirical analysis of each and every parameter in the form of Qubits of cryptographic analysis can be taken as measurement of the super-positions. This techniques of the crystal-kyber algorithms of the DSA of the algorithm to make the structural analysis to

secure the data from the malware attack in computing the data from the data-set.

[10,11,12] This can be statement regarding the status of the cryptographic report on the algorithm can be implemented in the structural analysis in the Quantum cryptographic strategies analysis in the dataset. The analysis can shows the few algorithm can provide the basic security regarding the attack an making the process optimize an flexible. The QNN approach can estimates the framework analysis which includes the various QNN algorithm to find out the structure hidden layer approach in authenticating the data in the data-set and make the error correction with the help of the activation key. [13,14,15] QNN noise resident factor analysis which can estimates the usage of the data scarcity and the data optimization and free from the noise intermediate usage of de-coherence techniques to remove the error of the hidden layer approach and can be updated in the calculation of the measurement of the super-position in terms of the complex integers. The QNN is the approach to find out the data in the form of the usage of manipulation of the data can be taken in the approach of hidden layer approach and make the computation fast and flexible. [16,17,18] This depict the strategical analysis of the learning model of the Quantum analysis in the cryptographic that can be taken in the form of the Qubits in the cryptographic approach , that can be more of the optimization of the data and the algorithm that can be taken in the terms of the neural network , cellular neural network that can work to remove the error from the activation functions.[19,20,21] The structural cryptographic algorithm that can be implemented in the terms of the data authentication and the data authorization of the data in the terms of the Quantum Qubits form in the 0 and 1. The process of the strategies analysis of the ensuring the algorithm can b e provide and the security and the malware attack analysis in the computation of the data-set. The function of the categorization of the data in terms of the DSA algorithm and the hash algorithm that provide the flexibility of the computation flexibility. [22,23,24] The cryptographic algorithm can be terms of the symmetric and the asymmetric algorithm that can be implemented in the terms of the data security attack and make the data free from the malware attack in the computations of the data analysis. The process of making the key for the authentication authorization of the data and male the data in the standardized form of computation. The algorithm are required in the cryptographic analysis that can be taken like DSA ,RSA CRYSTAL-Dilithium etc. used to be implemented for the Quantum cryptography analysis. [25,26,27] The Quantum classical Qubits is the form of the approach by which it can be taken as the form of digit 0 and 1 in the form of the continuous data states and can be implemented along with the complex integer values of the data computation. The processing techniques by which it can initialized with the value in the form of the 0 and 1. The process of analogous series can be computed and make the data optimization and the data scarcity.[28,29,30] The data can be taken in the form of the qubits in the form of binary computation and can be measured and make polarized by the terms of the 0 and 1. This can betaken in the terms of the data analysis in the shors and the Grovers algorithm that can be implemented with help of the operator and make the computation based on the complex computation integer of the super-position of the outcome.[31,32,33,34] The structural data analysis of the computation of the shors and the grovers algorithm and make the computation based on the QNN algorithmic approach. The data-set can be taken in the form the different parameter of the quantum approach in the molecular approach in detection the parameter in the form of the Qubits molecules in QM9.

III. Methodology

The NIST Algorithmic approach in quantum Learning is the phenomenon approach that can be secure and make the optimization of the data scarcity. It followed up the steps to in the classical Qubit forms that indicates the QNN learning and Shor;s and Grover's Algorithmic approach to secure and communicate the data and free from the attack analysis[4,5,6].

Step 1 : Initializing the dataset and making the data-structure analysis in the data-modelling in network processing Crystal-Kyber[21,22].

Step 2 : It estimates the classical structural analysis of data-modelling and the pre-processing techniques to optimizing the classical Qubits and the hidden layer data error detection [23].

Step 3: It is used to formulates the data modulated in the data approach and the quantum cryptography approach [30].

Step 4: The QNN and the Shor's and Grover's approach in the hidden layer approach to estimate the data analysis. It also initializing the data pf qubits to solve the error detection [24,25].

Step 5: The process of framework analysis processed until it remove all the data error detection and optimizing the data and the flexibility in the quantum cryptography [26,27].

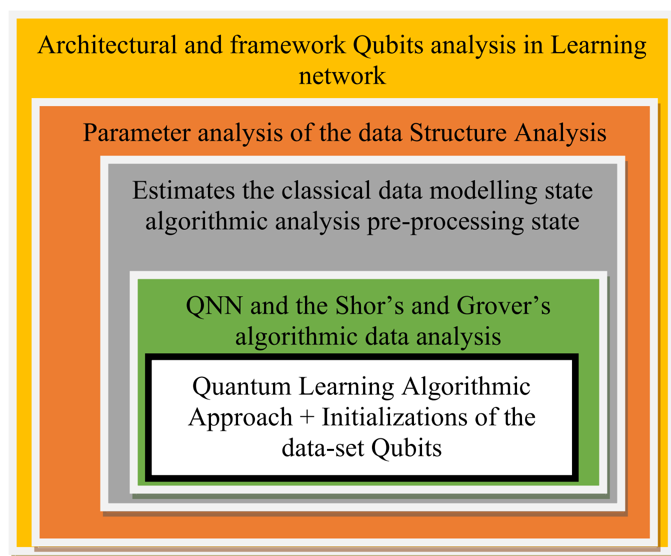

Fig. 2. Architectural framework analysis

A. *Classical Qubit Analysis*

The word classical Qubit in term of the quantum cryptography can be taken and the term of the quantum learning stature that can estimate the procedural analysis of the flexibility and the optimization and the data scarcity in the calculation of each and every parameter of the quantum learning Algorithmic approach[18]. The Qubit can be in 0 and 1 form to find the problem statement of the compliably super-position in equation 6,7[28,29].

$|€|= D|0> + E|1>$ …………(6) (D,E- complex integer)

This can estimates the strategies of the data in the form of Qubit states can be i.e.

$|€> =(1/\sqrt{3}|0> + (1/\sqrt{3})|1>$ …….(7) this process requires the modulation od state in calculation the every bits of analysis is [19]:

$|0> = |1/\sqrt{3}|^2 = 1/3$

$|1> = |1/\sqrt{3}|^2 = 1/3$

This type of computation can be taken in the form of the states in the form of the Qubit Complex number and can be taken in the form of 0 and 1 and recognizing in the form of the super position [31,32].

The cryptographic algorithm in the QNN approach can be taken as in the form of :

$A=\sum b1^2(z1+1) +\sum c1^2(z1, z1+1) +\sum d1^2(P+1)$………….(8)

$\partial/\partial t(b)= -q(z1,p1,Q)$……………………………………(9)

These two equation 8,9 can taken in the form of Quantum Qubit for in the terms of 0 and 1. The range can be taken in the polarized form between the -1 to 1. The two ranges make the molecular detection in terms of the hidden layer approach and the data optimization and the data scarcity.

B. *Flowchart Analysis*

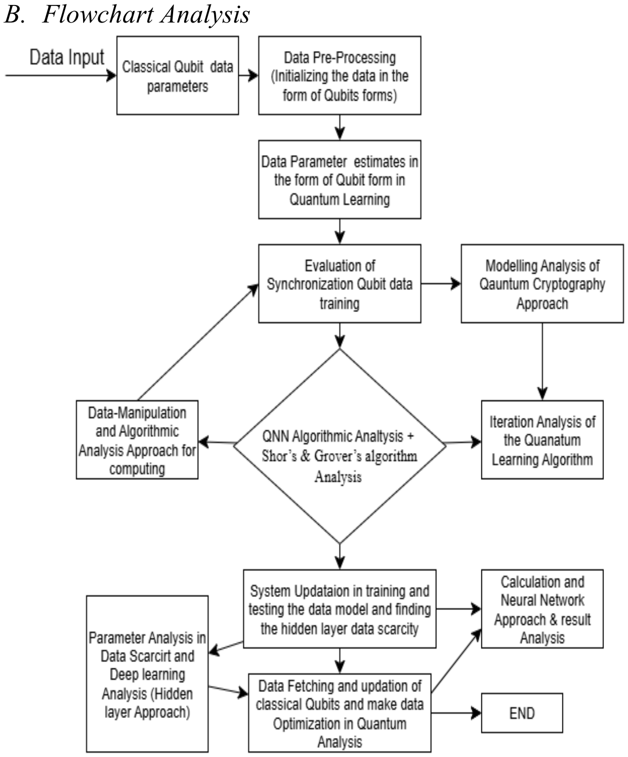

Fig. 3. Flowchart Analysis of Cryptographic Approach

IV. RESULT AND DISCUSSION

Dataset [33] [34] : It describe the estimation in the Quantum machine and the platform it is taken in the QISKIT machine Learning Ecosystem mainly used for the Quantum Learning Algorithm Analysis. The data can be taken in the form of the molecular state to estimate the prediction of structural analysis of each parameter in data scarcity and optimization. In the train and test the data can be taken in the dipole and the magnetic form of the molecular and the data can be taken in the form of the scaler and the potential energy strategies of the data-set. After the text edit of the MS Word Formatting toolbar. The Quantum Machine9-QM9 is the data that can be taken in the form of train and test strategies and it has different parameter of molecular around 17 and each value can be taken in the form of the x,y,z as an input of molecules in QM9 [34].

```
Fold 0
Training until validation scores don't improve for 100 rounds.
[100]	training's l1: 2.95863	valid_1's l1: 2.98979
[200]	training's l1: 2.88429	valid_1's l1: 2.94164
[300]	training's l1: 2.84503	valid_1's l1: 2.91957
[400]	training's l1: 2.8158	valid_1's l1: 2.90523
[500]	training's l1: 2.7896	valid_1's l1: 2.89082
Did not meet early stopping. Best iteration is:
[500]	training's l1: 2.7896	valid_1's l1: 2.89082
Fold 1
Training until validation scores don't improve for 100 rounds.
[100]	training's l1: 2.94631	valid_1's l1: 2.98034
[200]	training's l1: 2.87643	valid_1's l1: 2.93203
[300]	training's l1: 2.83604	valid_1's l1: 2.90949
[400]	training's l1: 2.811	valid_1's l1: 2.89866
[500]	training's l1: 2.79157	valid_1's l1: 2.89147
Did not meet early stopping. Best iteration is:
[500]	training's l1: 2.79157	valid_1's l1: 2.89147
Fold 2
Training until validation scores don't improve for 100 rounds.
[100]	training's l1: 2.90947	valid_1's l1: 2.93997
[200]	training's l1: 2.84248	valid_1's l1: 2.89605
[300]	training's l1: 2.79446	valid_1's l1: 2.86525
[400]	training's l1: 2.76997	valid_1's l1: 2.85437
[500]	training's l1: 2.75283	valid_1's l1: 2.84895
Did not meet early stopping. Best iteration is:
[500]	training's l1: 2.75283	valid_1's l1: 2.84895
```

Fig. 4. Train and Testing Analysis

```
meta_cols = ['id', 'molecule_name', 'atom_index_0', 'atom_index_1', 'scalar_coupling_constant']
target = 'scalar_coupling_constant'

fi = pd.DataFrame()
models = []

setSeeds(1234)

data_filt = data_train.drop(meta_cols, axis=1, errors='ignore')
predictions = np.zeros(len(data_filt))

kf = KFold(n_splits=FOLDS_VALID, random_state=1234, shuffle=True)
fold_splits = kf.split(data_filt)

for i, (dev_index, val_index) in enumerate(fold_splits):

    print('Fold', i)

    Xt, Xv = data_filt.loc[dev_index, :], data_filt.loc[val_index, :]
    yt, yv = data_train.loc[dev_index, target], data_train.loc[val_index, target]

    d_train = lgb.Dataset(Xt, yt)
    d_valid = lgb.Dataset(Xv, yv)

    watchlist = [d_train, d_valid]
    model = lgb.train(params_lgb,
                      train_set=d_train,
                      num_boost_round=500,
                      valid_sets=watchlist,
                      verbose_eval=100,
                      early_stopping_rounds=100)
```

Fig. 5. Molecular Qubit Analysis

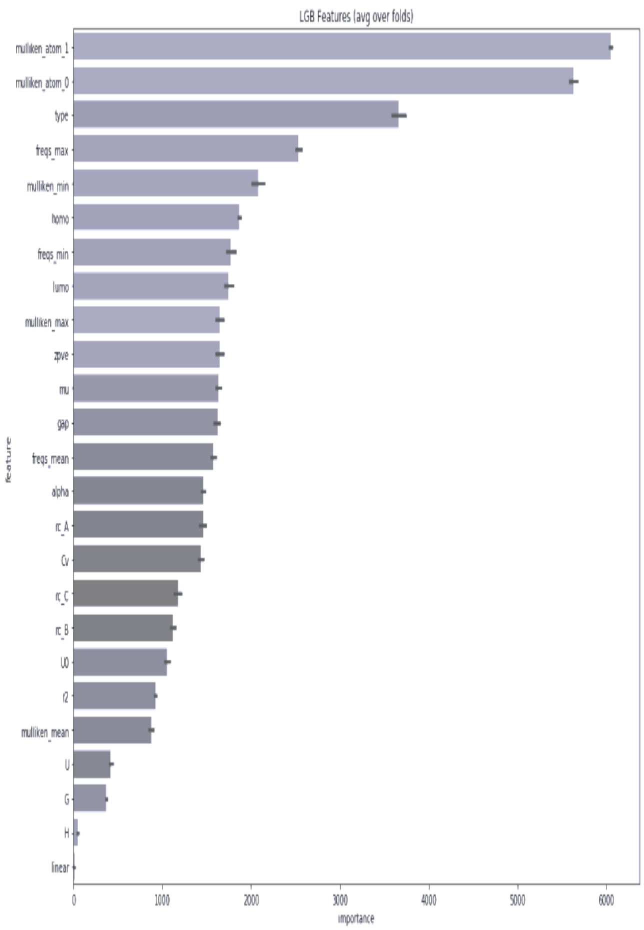

Fig. 6. Categorization of each parameter

V. CONCLUSION

The NIST Post-Quantum approach in the Quantum learning is describe the parameter analysis in the Quantum Learning approach in estimation and finding the error analysis through the QNN learning Algorithmic approach to strategies the classical Qubit measurement analysis in the super-position. The dataset can be in the form of molecular state in which finds the quantum measurement analysis in the QM9 [33,34] specify the analysis of the certain parameter. It takes 70%of training and 20% of training and the 10 % of the validation formation to predict the modulation of the dataset to predict the certain rotation of the molecular state. The process of Qubit learning of the classical to optimize the data and also reduction of the error analysis in computing the data in data-set. This process also estimate the structural analysis in the form the activation function that reduce the structure of error in hidden layer approach. It can mold and analysis in computing . In computation of each and every modulation of the data that can check and analysis the Qubit layer by layer and then compute in one frame of structure and produce the data scarcity and the optimization of the data analysis in quantum learning approach.

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